VOL. 00

Index and Study Guide

Library map, study tracks and how to use every volume

Source	The 35 source sites listed per volume in /plan
Generated	2026-10-10
Level	All levels
Estimated study time	2 hours
Language	English
Volume	Vol. 00 - Index and Study Guide

This volume is personal study material. The notes, examples, exercises and tests were written from scratch while following the source site's learning path. No content is affiliated with, endorsed by, or copied from the source site or the technology's owner.

Table of contents

About this library . . . 3

1.1 Summary . . . 3

1.2 The learning contract . . . 3

Volume index . . . 3

Study tracks . . . 5

3.1 Track A - Front end web . . . 5

3.2 Track B - Back end . . . 5

3.3 Track C - Languages and fundamentals . . . 6

3.4 Track D - Mobile . . . 6

3.5 Track E - DevOps and cloud . . . 6

3.6 Track F - Security and Web3 . . . 6

How to study a chapter . . . 7

4.1 Spaced repetition . . . 7

How to run the code . . . 7

5.1 Worked example of the loop . . . 7

Common mistakes that apply to all volumes . . . 8

Practice project ideas . . . 8

Mini test . . . 8

8.1 Answer key and explanations . . . 9

Cheat sheet . . . 9

Sources . . . 10

About this library

This is a library of 36 study volumes (00 + 35 technologies). Every volume is built from a free public tutorial or documentation site, following the same structure so you always know where to find things.

The notes in each volume were written from scratch while working through the source site's learning path: the summaries, explanations, exercises, tests and code examples are original. The source sites are only used as a syllabus, and are cited in each volume's Sources chapter.

1.1 Summary

- One volume per technology, plus this index and study guide.
- Every volume has the same skeleton, so navigation is always the same.
- Each chapter ends with practice exercises (with solutions) and a mini test.
- Each volume ends with a cheat sheet and a final exam with an answer key.
- All content is in English and is original writing, not copied text.
- Code examples are written and tested by the author, with expected output.

1.2 The learning contract

In every chapter	What it is for
Summary bullets	Recall the whole chapter in two minutes
Concepts + diagrams	Understand, not just memorise
Runnable examples	See the idea working
Mistakes and best practices	Avoid the classic traps
Exercises + solutions	Prove you can do it yourself
Mini test + key	Check yourself immediately

At the end of every volume	What it is for
Cheat sheet	Quick reference while working
Final exam	Measure the whole volume
Sources	Verify and go deeper

Volume index

Vol	Technology	Source site	Level	Est. hours
01	HTML	html.com	Beginner	20
02	CSS	web.dev/learn/css	Beginner	30
03	JavaScript	javascript.info	Beginner	40
04	React	reactplay.io	Intermediate	30
05	Vue	learnvue.co	Beginner	25
06	Angular	angular.dev/tutorials	Intermediate	30
07	Git	git-scm.com/book	All	15
08	Web3	learnweb3.io	Intermediate	25
09	Python	learnpython.org	Beginner	25
10	SQL	w3schools.com/sql	Beginner	20
11	Blockchain	cryptozombies.io	Intermediate	25
12	Next.js	nextjs.org/learn	Intermediate	25
13	AI (basics)	elementsofai.com	Beginner	20
14	PHP	phptherightway.com	Intermediate	25
15	API	rapidapi.com/learn	Beginner	15
16	Go	learn-golang.org	Beginner	20
17	Rust	rust-lang.org/learn	Intermediate	35
18	Design Patterns	refactoring.guru	Intermediate	30
19	TypeScript	typescriptlang.org/docs	Intermediate	25
20	C++	cplusplus.com/doc/tutorial	Intermediate	35
21	Java	docs.oracle.com/javase/tutorial	Intermediate	35
22	C#	dotnet.microsoft.com/learn/csharp	Beginner	30
23	Swift	swift.org/learn	Beginner	25
24	Django	djangoproject.com/start	Intermediate	25
25	Flask	flask.palletsprojects.com/tutorial	Beginner	15
26	Docker	docker.com/get-started	Beginner	15
27	Kubernetes	kubernetes.io/docs/tutorials	Intermediate	25
28	Linux	linuxjourney.com	Beginner	20
29	Cybersecurity	tryhackme.com	Beginner	30
30	DevOps	roadmap.sh/devops	Intermediate	25

Vol	Technology	Source site	Level	Est. hours
31	Cloud (AWS)	aws.amazon.com/training	Intermediate	30
32	Cloud (GCP)	cloudskillsboost.google	Beginner	30
33	Cloud (Azure)	microsoft.com/learn/azure	Beginner	30
34	Data Structures	visualgo.net	Intermediate	30
35	System Design	systemdesign.one	Advanced	30

Note. The estimated hours are for reading, doing every exercise and passing the final exam, not for skimming.

Study tracks

You do not have to read the volumes in order. Pick a track and follow it; each track is a short path through the library.

3.1 Track A - Front end web

Order	Volume	Why here
1	01 HTML	Page structure
2	02 CSS	Styling and layout
3	03 JavaScript	Behaviour and logic
4	19 TypeScript	Types at scale
5	04 React or 05 Vue	A component framework
6	12 Next.js	Production framework
7	15 API	Talking to servers

3.2 Track B - Back end

Order	Volume	Why here
1	09 Python	First server language
2	10 SQL	Data modelling and queries
3	25 Flask	Small services first
4	24 Django	Full applications
5	07 Git	Version control

Order	Volume	Why here
6	15 API	Designing endpoints
7	26 Docker	Packaging and shipping

3.3 Track C - Languages and fundamentals

Order	Volume	Why here
1	09 Python	The friendly first language
2	34 Data Structures	How to think about data
3	16 Go or 20 C++	A compiled language
4	17 Rust	Memory safety and speed
5	18 Design Patterns	Naming good solutions
6	35 System Design	Scaling what you build

3.4 Track D - Mobile

Order	Volume	Why here
1	03 JavaScript	Shared language
2	04 React	Component model
3	23 Swift	Native iOS basics
4	15 API	Backend communication

3.5 Track E - DevOps and cloud

Order	Volume	Why here
1	28 Linux	The server's language
2	07 Git	Source of truth
3	26 Docker	Containers
4	27 Kubernetes	Orchestration
5	30 DevOps	The culture and pipeline
6	31/32/33	Pick one cloud

3.6 Track F - Security and Web3

Order	Volume	Why here
1	29 Cybersecurity	Security mindset
2	28 Linux	Systems knowledge
3	03 JavaScript	Tooling
4	11 Blockchain	Solidity and EVM
5	08 Web3	Decentralised applications

How to study a chapter

Follow this loop for every chapter, in every volume. It takes about 60 to 90 minutes per chapter and it works because you are forced to produce, not consume.

1. Read the summary first (2 min). Decide if you already know the material.
2. Read the concepts once (10 min). Do not code yet.
3. Run every example (15 min). Type it yourself, never copy and paste.
4. Close the book and write the example from memory (10 min).
5. Do the exercises (20 min). Look at the solutions only after trying.
6. Take the mini test immediately (10 min). Score yourself with the key.
7. Anything you missed goes on a review list. Re-read it the next day.

> Tip. The single biggest predictor of success is step 6: testing yourself immediately. It feels worse than re-reading and works far better.

4.1 Spaced repetition

Review each chapter twice: once the next day and once after one week. Two fast reviews beat one long re-read. A review is: read the summary bullets, then redo the mini test from memory.

How to run the code

Every code block is numbered so explanations can refer to specific lines. The examples are minimal on purpose.

- They are self-contained and need no frameworks unless a chapter says so.
- The expected output is always shown under the example.
- If the output differs, read the code line by line before debugging.

5.1 Worked example of the loop

Python

```python
# 1  A minimal example, numbered lines, expected output below
# 2  Compute the average of a list and handle the empty case.
def average(values):
    if not values:
        return None
    return sum(values) / len(values)

print(average([10, 20, 30]))
print(average([]))
```

output

```
20.0
None
```

Step 4 means you should be able to write that `average` function, from memory, in under two minutes.

Common mistakes that apply to all volumes

Mistake	Why it hurts	Fix
Reading without running code	False confidence	Type every example
Looking at solutions too early	You learn nothing	Struggle for 10 min first
Skipping the mini tests	You never find the gaps	Always take them
Moving on with an unsolved bug	The gap compounds	Debug before continuing
Binge-reading chapters	Retention collapses	Two chapters a day, maximum
Never re-reading	Memory decays	Review next day + 1 week

Practice project ideas

Each project combines volumes from a track. Build one after finishing a track.

1. Personal site (01, 02, 03): responsive, accessible, deployed for free.
2. Task app with an API (03, 09 or 24, 10): auth, CRUD, database.
3. Weather dashboard (04, 15): fetch a public API, cache, error states.
4. Containerised service (09, 26, 27): Docker image plus a deployment.
5. URL shortener with analytics (09, 10, 26, 30): full pipeline.
6. Design-pattern refactor (18, 09): take messy code and improve it.

Mini test

1. Multiple choice: what is the recommended maximum number of new chapters to study per day? - a) as many as possible - b) two - c) ten - d) it does not matter
2. True or false: you should copy and paste every code example to save time.
3. Multiple choice: which step in the study loop gives the biggest recall boost? - a) reading the concepts twice - b) watching a video first - c) taking the mini test immediately - d) reading the summary
4. True or false: the summaries exist so you can skip the chapter.
5. Ordering: put these in the right order: do exercises, read summary, take mini test, run examples.
6. Short answer: what should you do the day after finishing a chapter?
7. Multiple choice: which track starts with 01 HTML and 02 CSS? - a) Back end - b) Front end - c) DevOps - d) Security

8.1 Answer key and explanations

1. b - two chapters a day keeps retention high; bingeing collapses it.
2. False - typing examples yourself forces you to read every character.
3. c - immediate retrieval practice beats re-reading.
4. False - they are for recall and review, not a substitute for the chapter.
5. read summary, run examples, do exercises, take mini test - that is the loop.
6. Review it - read the summary bullets and redo the mini test, then again after one week.
7. b - Track A (front end) starts with HTML and CSS.

Cheat sheet

Task	Where
Find a volume	Volume index above
Start from zero	Track A or Track B
Learn a language	Track C
Learn the cloud	Track E
Learn security	Track F

Study step	Time
Summary bullets	2 min
Concepts	10 min
Run + retype examples	25 min

Study step	Time
Exercises	20 min
Mini test	10 min

Rule of thumb	Why
Test yourself immediately	Best recall
Two reviews: next day, one week	Beats re-reading
Type, never paste	Forces attention
Debug before moving on	Gaps compound

Sources

- The 35 source sites listed in the volume index, one per technology.
- Per-volume detailed source lists live in each volume's final chapter.